

PA6.6 – Polyamide 6.6 PA66 FR

AKROMID® A3 1 FR black (3051)

Tensile modulus

3500 MPa

1 mm/min

ISO 527-2

Stress at yield

83 MPa

50 mm/min

ISO 527-2

Charpy impact strength

70 kJ/m²

23°C

ISO 179-1/1eU

AKROMID® A3 1 FR black (3051) is an unreinforced, heat stabilised, halogen- free flame retardant polyamide 6.6

Typical applications

Switch parts and housings in the electric industry

**Mechanical Properties**

Tensile modulus (1 mm/min | ISO 527-2)

d.a.m.

3500 MPa

conditioned

1500 MPa

Stress at yield (50 mm/min | ISO 527-2)

d.a.m.

83 MPa

conditioned

55 MPa

Strain at break (50 mm/min | ISO 527-2)

d.a.m.

> 5 %

conditioned

>20 %

Charpy impact strength (23°C | ISO 179-1/1eU)

d.a.m.

70 kJ/m²

conditioned

n.b.

Charpy notched impact strength (23°C | ISO 179-1/1eA)

d.a.m.

4,5 kJ/m²

conditioned

7 kJ/m²**Thermal Properties**

Temperature of deflection under load HDT/A (1,8 MPa | ISO 75)

80 °C

Temperature of deflection under load HDT/B (0,45 MPa | ISO 75)

220 °C

Melting temperature (DSC, 10K/min | DIN EN ISO 11357-3)

260 °C

Disclaimer:

All specifications and information given on this website are based on our current knowledge and experience. A legally binding promise of certain characteristics or suitability for a concrete individual case cannot be derived from this information. The information supplied here is not intended to release processors and users from the responsibility of carrying out their own tests and inspections in each concrete individual case. AKRO®, AKROMID®, AKROLEN®, AKROLOY®, AKROTEK®, ICX® and PRECITE® are registered trademarks of the Feddersen Group.

**Flammability**

Burning rate (UL 94)	
0,4mm Wall thickness	V-0 Class
0,8mm Wall thickness	V-0 Class
1,6mm Wall thickness	V-0 Class
3,2mm Wall thickness	V-0 Class
GWFI (IEC 60695-2-12)	
0,4mm Wall thickness	960 °C
0,8mm Wall thickness	960 °C
1,6mm Wall thickness	960 °C
3,2mm Wall thickness	960 °C
GWIT (IEC 60695-2-13)	
0,4mm Wall thickness	775 °C
0,8mm Wall thickness	750 °C
1,6mm Wall thickness	750 °C
3,2mm Wall thickness	775 °C
HWI (UL 746A)	
0,8mm Wall thickness	4 PLC
HAI (UL 746A)	
0,8mm Wall thickness	0 PLC
Oxygen index (ISO 4589-2)	> 32 %
Burning rate (<100 mm/min) (> 1 mm Thickness FMVSS 302)	+

**General properties**

Density (23°C ISO 1183)	1,18 g/cm ³
Molding shrinkage (flow ISO 294-4)	1,3-1,6 %
Molding shrinkage (transverse ISO 294-4)	1,3-1,6 %

**Electrical Properties**

Comparative tracking index (Test liquid A IEC 60112)	>600 V

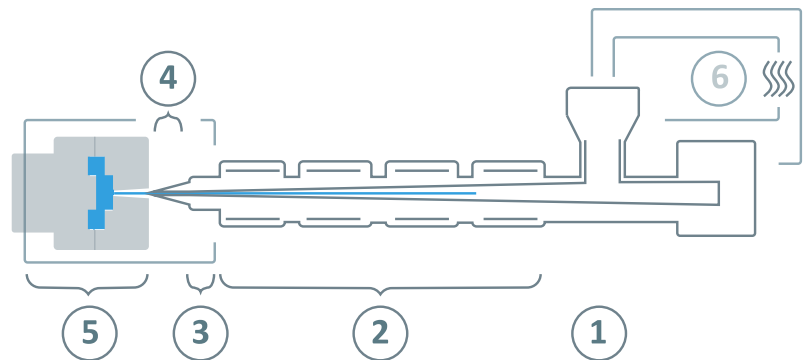
Disclaimer:

All specifications and information given on this website are based on our current knowledge and experience. A legally binding promise of certain characteristics or suitability for a concrete individual case cannot be derived from this information. The information supplied here is not intended to release processors and users from the responsibility of carrying out their own tests and inspections in each concrete individual case. AKRO®, AKROMID®, AKROLEN®, AKROLOY®, AKROTEK®, ICX® and PRECITE® are registered trademarks of the Feddersen Group.

PA6.6 – Polyamide 6.6 PA66 FR

AKROMID® A3 1 FR black (3051)**Processing information**

The listed values are recommendations. Higher values should be used for higher glass loadings. We recommend only dehumidifying or vacuum dryers. Extensive drying can cause filling problems and surface defects.



⑥	Drying time	2 - 4 h
	Drying temperature ($\tau \leq -30^{\circ}\text{C}$)	80°C
	Processing moisture	0,02 - 0,08%
①	Feed section	60 - 80°C
②	Temperature zone 1 - Zone 4	260 - 290°C
③	Nozzle temperature	260 - 300°C
④	Melt temperature	270 - 290°C
⑤	Mold temperature	60 - 80°C
→	Holding pressure, spec.	300 - 800 bar
←	Back pressure, spec.	30 - 100 bar
	Injection speed	medium
	Screw speed	5 - 10 m/min

Disclaimer:

All specifications and information given on this website are based on our current knowledge and experience. A legally binding promise of certain characteristics or suitability for a concrete individual case cannot be derived from this information. The information supplied here is not intended to release processors and users from the responsibility of carrying out their own tests and inspections in each concrete individual case. AKRO®, AKROMID®, AKROLEN®, AKROLOY®, AKROTEK®, ICX® and PRECITE® are registered trademarks of the Feddersen Group.